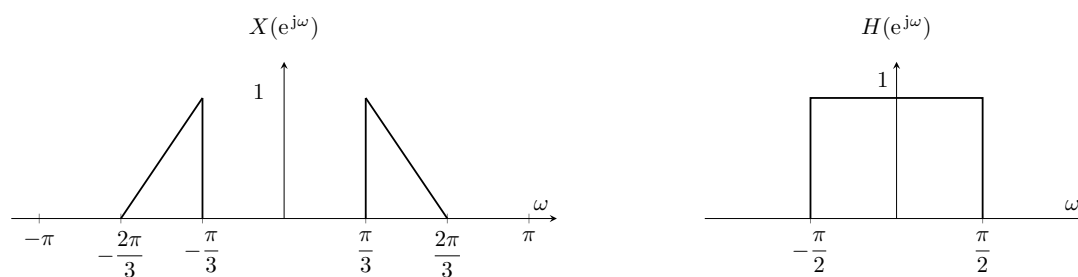


Homework 2: Multirate Signal Processing

1 Multirate System Analysis

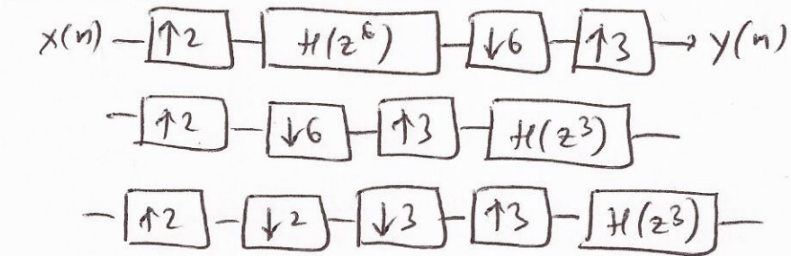
Consider the following multirate system:



- Express $Y(z)$ in terms of $X(z)$ and $H(z)$.
- Sketch $|Y(e^{j\omega})|$.

Solution. See next page

a.

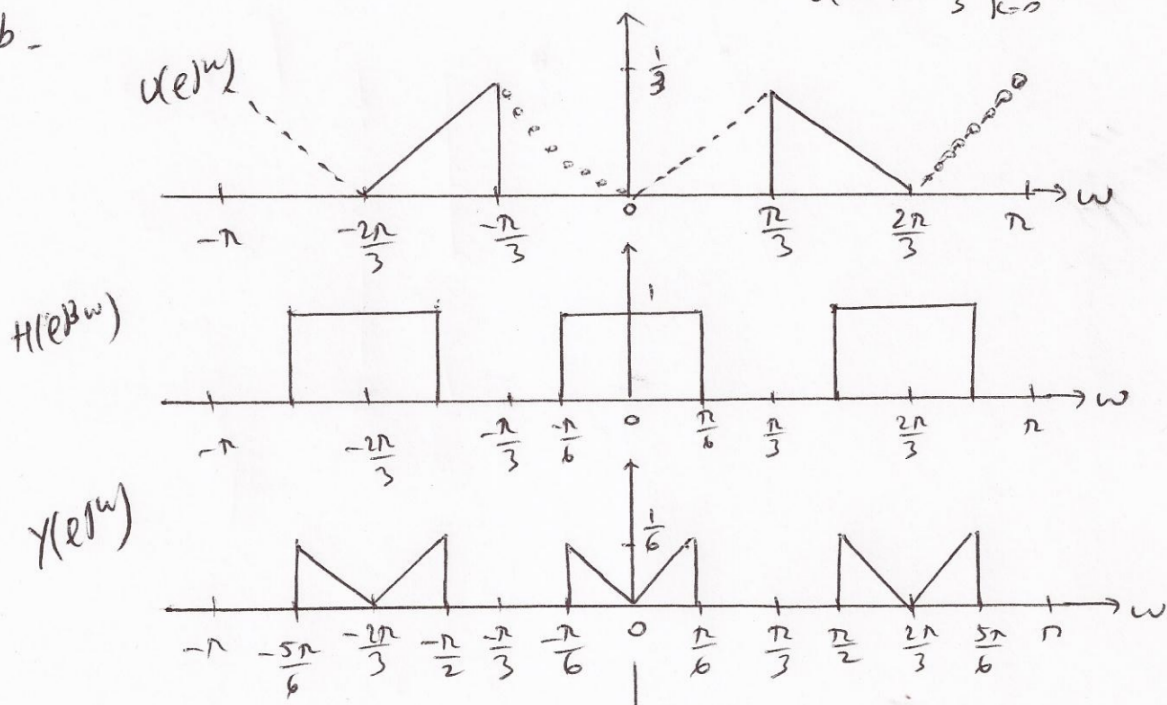
 $x(z)$

$$\frac{1}{3} \sum_{k=0}^2 x(z e^{j\frac{2k\pi}{3}})$$

$$y(z) = \frac{1}{3} H(z^3) \sum_{k=0}^2 x(z e^{j\frac{2k\pi}{3}})$$

$$U(e^{j\omega}) = \frac{1}{3} \sum_{k=0}^2 X(e^{j(\omega - \frac{2k\pi}{3})})$$

b.



2 Polyphase Representations

Suppose we want to decompose $H(z)$ into $M = 4$ polyphase components, i.e., we want to find E_0, E_1, E_2 , and E_3 such that

$$H(z) = E_0(z^4) + z^{-1}E_1(z^4) + z^{-2}E_2(z^4) + z^{-3}E_3(z^4).$$

- (a) Prove that the decomposition into the polyphase components is unique.

Solution. There are many possible proofs of this claim. An easy one based on the lecture material is to recall from lecture 6 that $H(z)$ is related to the polyphases via an unnormalized 4-point DFT matrix the invertibility of this matrix is sufficient to guarantee that the polyphase decomposition is unique.

An alternative proof would be to expand out $H(z)$ via its z -transform and use the fact that the four polyphases partition the integers (the time indices) into the residue classes modulo 4, which is a unique decomposition of the integers.

- (b) What polynomial multiplies $(1 - az^{-1})$ to produce $(1 - a^4z^{-4})$?

Solution. Multiply by $G(z) = (1 + az^{-1} + a^2z^{-2} + a^3z^{-3})$. The way you can find this is to observe that the ratio

$$\frac{1 - a^4z^{-4}}{1 - az^{-1}} \quad (2.1)$$

is the closed form expression of a truncated geometric series and that $G(z)$ is that truncated geometric series.

- (c) Determine the 4 polyphase components of $\frac{1}{1-az^{-1}}$.

Solution. By the same computations for $M = 3$ in lecture, we have that

$$E_k(z) = \frac{a^k}{1 - a^4z^{-1}}, \quad k = 0, 1, 2, 3. \quad (2.2)$$

- (d) Let $H(z) = \frac{1+2z^{-1}+5z^{-2}}{1-(5/6)z^{-1}+(1/6)z^{-2}}$. Determine the 4 polyphase components.

Solution. The partial fractions decomposition is

$$30 + \frac{75}{1 - \frac{1}{2}z^{-1}} - \frac{104}{1 - \frac{1}{3}z^{-1}}. \quad (2.3)$$

Observe that the 4 polyphases of 30 are $(30, 0, 0, 0)$ (since 30 in the z -domain corresponds to $30\delta[n]$ in the time domain). Using the identity from (c), we have that

$$E_k(z) = 30 \cdot \mathbf{1}\{k = 0\} + \frac{75(1/2)^k}{1 - \frac{1}{16}z^{-1}} - \frac{104(1/3)^k}{1 - \frac{1}{81}z^{-1}}, \quad k = 0, 1, 2, 3, \quad (2.4)$$

where $\mathbf{1}\{k = 0\}$ is the indicator of $\{k = 0\}$, i.e., it is 1 when $k = 0$ and 0 otherwise.

3 Perfect-Reconstruction (PR) Filter Banks

Let (H_0, H_1) denote the analysis filters and (G_0, G_1) denote the synthesis filters of a two-channel PR filter bank.

- (a) Interchange H_k and G_k , $k = 1, 2$, i.e., use (G_0, G_1) in the analysis bank and (H_0, H_1) in the synthesis bank. Is this new system PR?

Solution. We have that

$$\begin{aligned} G_0(z)H_0(z) + G_1(z)H_1(z) &= 2z^{-L} \quad L \text{ is odd} \\ G_0(z)H_0(-z) + G_1(z)H_1(-z) &= 0. \end{aligned}$$

If we interchange G_k with H_k , the resulting system $\hat{H}_k := G_k$ and $\hat{G}_k := H_k$ is still PR since

$$\begin{aligned} \hat{G}_0(z)\hat{H}_0(z) + \hat{G}_1(z)\hat{H}_1(z) &= H_0(z)G_0(z) + H_1(z)G_1(z) = 2z^{-L} \\ \hat{G}_0(z)\hat{H}_0(-z) + \hat{G}_1(z)\hat{H}_1(-z) &= H_0(z)G_0(-z) + H_1(z)G_1(-z) = 0. \end{aligned}$$

- (b) Define another system via $\tilde{H}_k(z) = H_k(-z)$ and $\tilde{G}_k(z) = G_k(-z)$, $k = 1, 2$. Is the the new system PR?

Solution. We have that

$$\begin{aligned} \tilde{G}_0(z)\tilde{H}_0(z) + \tilde{G}_1(z)\tilde{H}_1(z) &= G_0(-z)H_0(-z) + G_1(-z)H_1(-z) = -2z^{-L} \\ \tilde{G}_0(z)\tilde{H}_0(-z) + \tilde{G}_1(z)\tilde{H}_1(-z) &= G_0(-z)H_0(z) + G_1(-z)H_1(z) = 0. \end{aligned}$$

So strictly speaking the system is not PR since it has a constant scaling (constant being -1) in this case, but if we additionally postprocess the output (by multiplying by -1) it would be PR.

- (c) Suppose that $H_1(z) = H_0(-z)$ and that H_0 is FIR. Does there exist a PR system that satisfies these constraints? If yes, give the general form for the filters of such a system. If no, prove that no such PR system exist.

Solution. See the lecture 8 notes, where we solved this problem in class.